Crowd-Risk Forecasting for Safer Large Events in Seoul

Team Members: Jimmy Kim, Minhyuk Park

Project Goal and Motivation

In 2022, the Itaewon Halloween tragedy showed how quickly a crowded area can become dangerous. A close friend of my parents passed away during that incident, which motivated this project idea. Large seasonal events and nightlife areas in Seoul can become crowded very fast. When too many people gather in a small space, serious safety risks can appear. Our project will build a simple forecasting tool that estimates how crowded a place may be at a specific time. This tool can help people plan earlier and make safer choices, such as choosing better times or preparing more staff. We aim to use past crowd patterns to provide clear guidance, especially during busy periods like Halloween weekends.

Data

We will use Seoul living population data as our main dataset. Our main target variable will be the administrative dong level living population for Korean residents. We will predict the living population count for a given region and time. We will create basic time based features such as hour of day, day of week, and month. If useful, we will include additional signals. These include Seoul culture event listings, which provide event locations and dates, and Seoul Metro transfer station passenger counts, which can act as a transit based crowd signal.

Planned Analysis and Why

We will build and compare two regression models. The first model will be Linear Regression, which serves as a simple baseline and is easy to understand. The second model will be a Random Forest Regressor, which can capture more complex patterns in the data. We will compare the results from both models. We will also discuss the trade off between models that are easy to explain and models that may produce better predictions.

Performance Measures

We will measure performance using RMSE and MAE on a held out test set. If time allows, we will also use cross validation. We will present error results using simple charts. We will also report feature importance from the Random Forest model.

Who Will Do What

- Jimmy Kim will handle data collection, data cleaning, feature engineering, exploratory data analysis, and documentation of data limits.
- Minhyuk Park will build the models, run evaluations using RMSE and MAE, conduct comparison experiments, and create result charts.
- Both team members will refine the project idea, interpret results, write the final report, and prepare presentation slides.

Timeline

- By Feb 11, we will download the living population data, select the final region unit, and define the exact prediction target. We will also create one clean table that matches region and time to the target value.

- By Feb 16, we will complete basic exploratory plots and train the baseline Linear Regression model. We will also prepare the three slide presentation with one figure explaining the problem and data.
- By Mar 05, we will train the Random Forest model, tune a small set of parameters, and compare both models using RMSE and MAE. We will also write the main results and include at least two comparison plots.
- By Mar 06, we will finish the final report and include the final model comparison, key feature findings, and main data limitations.